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CO₂ Foamed Acid Rheology for High-Temperature Acid Fracturing

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Abstract

Foamed acid fracturing is gaining importance to maximize flowback recovery and is particularly applicable when reservoir energy is not sufficient to effectively flow back the well. Foamed acid offers additional benefits, such as reduced water consumption and improved acid diversion, in addition to potential features like retardation and deeper conductivity generation. The thermal stability of foam at high temperatures is one of the main challenges. In this paper, the rheology of two CO₂ foamed HCl acid systems having retarded acid characteristics was studied in a high-temperature environment.

The development of foam viscosity depends on the type of gas, desired foam quality characteristics, the external phase fluid viscosity as a function of the type of acid blend and strength, corrosion inhibitor package, foaming agent, and stabilizers. The performance of CO₂ foamed acid systems was assessed by several laboratory tests, characterized by compatibility, corrosion testing and foam rheology. The rheological properties of different qualities of CO₂ foams containing two retarded acid systems were measured at a range of temperatures, i.e., 200 and 250°F, and shear rates from 100 to 1600 1/s, using a custom-designed foamed acid etching equipment which has both rheology and acid etching capability.

Acid fracturing has three key aspects for successful acid stimulation treatments: controlling acid reactivity, managing fluid loss, and generating conductivity. The overall acid reactivity, which governs the dissolution of carbonate rock, can be significantly decreased after foaming, even with the same acid strength and temperature. This work develops two base non-viscous acid systems for foaming applications: one based on a retarding agent and the other based on an organic acid, both combined with a strong mineral acid, HCl. The Modulated Retarded Acid (MRA) system, which can be formulated with different concentrations of hydrochloric acid and mineral acids, utilizes a gelling agent and exhibits a CO₂ foamed viscosity in the range of 80-105 cP at a shear rate of 100 1/s at 200°F. The second system, Low Viscosity Retarded Acid (LVRA), developed with organic acid, exhibits a viscosity in the range of 44-45 cP at a shear rate of 100 1/s at 250°F. The CO₂ foamed rheology of both the MRA and LVRA acid systems exhibits stable viscosity, showing potential for use in acid fracturing applications.

Introduction

The use of carbon dioxide and nitrogen as acidizing aids for recovering stimulation fluids and eliminating water or emulsion blocks was first demonstrated in the 1960s (Neill et al., 1964). In the 1970s, foamed acid was introduced as a means to improve acid retardation and manage leak-off during acid fracturing (Holcomb, 1977; Scherubel and Crowe, 1978; Briscoe, 1979). The following decade saw a significant increase in research, with numerous studies published in the 1980s that advanced our understanding of the rheology, reaction kinetics, leak-off, etched fracture conductivity, and field deployment of N₂ and CO₂ foamed acid systems (Ford et al., 1980; Ford, 1981; Ford and Roberts, 1985; Reidenbach et al., 1986; Anderson and Fredrickson, 1989).

The acid reactivity, fluid loss, and conductivity generation are three essential challenges of acid fracturing that must be addressed for acid stimulation treatments to be successful (Gdanski and Lee, 2005). Acid reactivity, which is influenced by the content and temperature of carbonate minerals, governs the dissolution rate of carbonate rocks. In acid fracturing, synthetic polymer gelled acids are used to regulate reactivity and mass transfer. However, maintaining the treating pressure above the fracture extension pressure is challenging due to significant acid fluid loss, resulting in a more aggressive acid treatment with a shorter etching time. To mitigate this, viscosifying the acid with additives such as polymers, surfactants, foams, and emulsions can help reduce fluid loss. Effective reactivity management and fluid loss control are crucial for achieving good conductivity generation. Furthermore, sufficient and uneven carbonate removal during acid fracturing can enhance conductivity creation.

Acid systems based on foamed hydrochloric acid (HCl) have shown significant improvements in stimulating fluid performance, particularly in addressing fluid loss management, reactivity control, and conductivity production. According to Ford and Roberts (1985), foaming HCl with nitrogen results in mass transfer or diffusion-controlled acid spending. Notably, foamed acid eliminates the need for polymers to regulate fluid loss, as observed by Ford (1981). A comparative study by Anderson and Fredrickson (1989) on non-foamed and foamed acid etched fracture conductivities for various acids and test settings revealed that foaming acid substantially increases fracture conductivities. The foaming process helps to wash away fines from the fracture face, enhancing the erosive effects and contributing to improved conductivity. A review of published literature by Karadkar et al. (2018) highlights the benefits of using foamed acid in acid fracturing, further supporting its effectiveness.

A liquid is classified as an energized fluid if it contains less than 52% gas, and as a foam if it contains more than 52% gas (Jacobs, 2014). Foamed fluids offer a promising alternative to traditional stimulation fluids in various situations, such as low reservoir pressure, water-sensitive formations, and cases where rapid cleanout is necessary. According to Hutchins and Miller (2003), the three primary characteristics of foam are its quality, texture, and rheology, which are essential factors in determining its behavior and performance. The following equation,

$$\Gamma = \frac{100V_g}{V_g + V_l} \quad (1)$$

where V_g is gas volume and V_l is liquid volume, is used to calculate foam quality (Γ), expressed as a percentage (%) at a certain temperature and pressure. The dispersed gas phase's bubble size distribution is referred to as the foam texture. The foam texture refers to the bubble size distribution of the dispersed gas phase, which qualitatively describes the distribution of small and large bubbles, as well as the homogeneity or heterogeneity of the mixture. Foams typically exhibit a systematic hexagonal pattern due to the dispersion of gas through a continuous surfactant solution (Belhajj et al., 2014). This pattern is a result of the surfactant's ability to reduce surface tension and facilitate the formation of a uniform bubble structure.

Rheological measurements of foam are crucial for evaluating its structural stability, optimizing foaming agent concentrations, and determining key engineering parameters, such as the behavior index (n') and consistency index (K'). These parameters play a vital role in fracturing simulators, where they are used to

estimate operating conditions and predict fracture geometry (Hutchins and Miller, 2003). The development of foam viscosity is a complex process that depends on several factors, including the type of gas, foam quality, and the viscosity of the external phase fluid, which is influenced by the type of acid blend and strength, corrosion inhibitor package, foaming agent, and foam stabilizer. Additionally, foam viscosity is also affected by operating conditions such as shear rate, temperature, and pressure.

Recently, low-viscosity retarded acid systems have been proposed as an alternative to traditional acid retardation methods, such as polymer addition or emulsification in diesel. For example, Sayed et al. (2018) developed a novel mixed acid system by combining an organic acid with a strong mineral acid. Similarly, Reyes and Beuterbaugh (2020) created a HCl acid-based fluid blended with a retarding agent, which exhibits desirable properties (Santos et al., 2022) including high dissolving capacity, strong multivalent ion stabilization in the acidic pH range, and high temperature stability. This study builds upon these developments by investigating the properties of these two retarded acid systems when foamed with CO₂ for high-temperature acid fracturing applications. The performance of these CO₂ foamed acid systems was assessed for compatibility, corrosion, and static foaming in one of our previous publication (Karadkar et al, 2023). In this paper, the rheology of two CO₂ foamed acid systems with retarded acid characteristics was examined in a high-temperature environment.

Experimental Methods

Materials

The acid blends were prepared using 31% HCl acid as the base. To protect steel from corrosion, corrosion inhibitors and intensifiers specifically designed for strong mineral acids were added. Acid retarding agents were also incorporated into the respective acid blends to reduce the reactivity of the acid while maintaining its capacity for dissolving rock. Additionally, other necessary additives were included, such as iron control agents, H₂S scavengers, foaming agents, and foam stabilizers, to ensure the optimal performance of the acid blends.

Corrosion testing

The performance of the corrosion inhibitor package was evaluated using a high-pressure, high-temperature (HPHT) autoclave. T-95 coupons with a surface area of 4.25 in² were used for the test. The autoclave was pressurized to 500 psi and subjected to a 30-minute heat-up period, followed by a 15-minute cool-down period. The corrosion weight loss was calculated and expressed in units of lb/ft², normalized to the total surface area of the coupons. The passing criteria for the corrosion test were set at a weight loss of 0.05 lb/ft² or less over a period of 6 hours.

Rheology measurement

The viscosity of CO₂ foam was measured using a custom-designed circulating-loop foam rheometer, which consists of a 40-foot-long helically coiled tube with an internal diameter of 0.15 inches. To ensure consistency and evaluate foam stability, the foam quality was maintained between 70% and 75% throughout the experiments. A schematic diagram of the apparatus is shown in Figure 1. Foam stability was assessed by applying a shear rate of approximately 1200 s⁻¹ using a recirculating pump, followed by a gradual reduction to 100 s⁻¹, while maintaining a constant foam quality. The foam viscosity was calculated as the ratio of shear stress to shear rate. The fluid was circulated through the coiled loop at a controlled shear rate, which was determined based on the measured flow rate using a Coriolis meter, and the shear stress was calculated from the pressure drop across the coil. The desired foam quality was achieved by injecting CO₂ into the liquid phase while maintaining a constant system pressure.

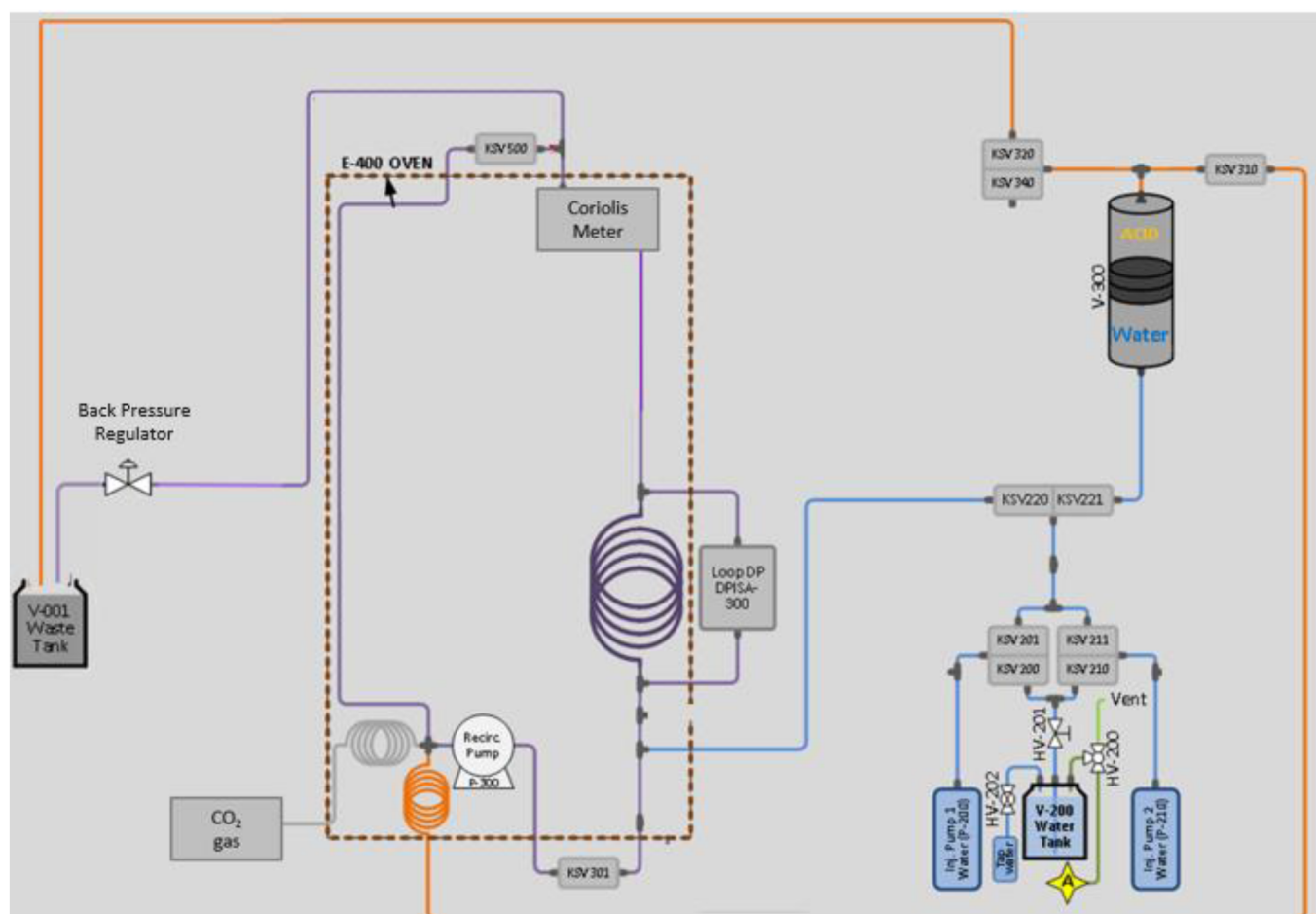


Figure 1—Schematic of foam rheometer.

Retarded acid systems

During acid fracturing, the use of strong mineral acids like HCl poses significant challenges, including corrosion issues with metals and limited penetration due to unfavorable reaction kinetics between the acid and rock matrix (Sayed et al., 2020). To mitigate these issues, the industry has employed emulsified acid, where diesel is used as an external phase emulsified with HCl. However, emulsified acid has its own drawbacks, including high friction losses and rigorous mixing requirements, which add complexity and make quality monitoring in the field difficult. Recently, low-viscosity retarded acid systems (LVRA) have been introduced as an alternative to acid retardation by polymer addition or emulsification in diesel. For example, Sayed et al. (2018) developed a novel mixed acid system by combining an organic acid with a potent mineral acid. Similarly, Reyes et al. (2020) created an HCl acid-based fluid blended with a retarding agent, which exhibits desirable characteristics such as high dissolving capacity, strong multivalent ion stabilization in the acidic pH range, and high temperature stability. In this study, the latter system is referred to as Modulated Retarded Acid (MRA), which can be formulated with different concentrations of hydrochloric acid. These two acid systems serve as the base fluids for the development of the CO₂ foaming application presented in this paper.

Foamed acid

During acid fracturing, acid is injected into a hydraulically created fracture, where it leaks into the formation, flows unevenly along the fracture, and reacts with the rock. After the hydraulic pressure has dissipated, the fracture closes, and conductivity is preserved through an etching process, in which relatively undissolved

portions of the rock serve as pillars, leaving more dissolved regions as open channels. The uneven etching of the fracture walls creates the conductivity after closure (Deng et al., 2012). However, in conventional acid fracturing, nitrogen lifting with coiled tubing is often required to initiate production at lower reservoir pressures. In contrast, foamed acid can eliminate the need for nitrogen lifting, providing efficient acid fracturing and faster spent acid flowback. Foamed acid is particularly beneficial for low-pressure, low-permeability, and liquid-sensitive formations, as well as those that produce a high volume of fines but are highly soluble. The advantages of foamed acid stimulation fluids include reduced liquid content, improved fluid loss control, built-in gas aid, minimal well shut-in time, and enhanced fracture conductivity (Ford et al., 1980). To create foamed acid with HCl as the base, various foaming agents can be used, such as sodium dodecyl sulfate, polyethylene glycol ethers of alkylated phenol, and mixtures of these with fluorinated alkyl quaternary ammonium iodide compounds (Holcomb, 1977). Figure 2 shows picture of foamed acid generated using CO₂ purging in base acid system.



Figure 2—CO₂ foamed acid system

Corrosion Testing

To prevent metal corrosion during acidizing operations, it is essential to use acid corrosion inhibitors. The selection of a corrosion inhibitor should be based on corrosion testing and its environmental impact. In this study, we utilized an autoclave to simulate high-pressure, high-temperature (HPHT) conditions and evaluated the corrosion performance of T-95 corrosion coupons. To assess the worst-case scenario, both acid systems were tested without CO₂ foaming. The results of the corrosion testing, including the test conditions, are presented in Table 1. Both base acid systems passed the corrosion tests for periods of 6 hours at temperatures of 300°F and 325°F. Notably, the Modulated Retarded Acid (MRA) acid service treatment fluid was formulated with the necessary additives to ensure foam stability and fluid compatibility.

Table 1—Corrosion performance testing non-foamed acid systems.

Test conditions	22% LVRA	15% MRA
Coupon Type	T-95	T-95
Test Duration (hr)	6	6
Test Temperature (°F)	250	325
Test Pressure (psi)	500	500
Corrosion Loss (lbm/ft ²)	0.019	0.026

Rheology measurement

The development of foam viscosity is influenced by several factors, including the type of gas, foam quality, external phase fluid viscosity, acid blend and strength, corrosion inhibitor package, foaming agent, foam stabilizer, shear rate, temperature, and pressure. The rheology of CO₂ foamed acid systems was investigated using a circulating foam rheometer at 200°F and 250°F. The viscosity of the CO₂ foam was measured using a circulating-loop foam rheometer, which consists of a 40-foot-long, helically-coiled tube with an inner diameter of 0.15 inches. To evaluate foam stability, a foam quality of 70-75% was maintained and adjusted. Foam stability was assessed by applying a shear rate of 1100-1300 1/s, while maintaining the foam quality, and then reducing it to 100 1/s to obtain a stable viscosity measurement. The viscosity of the foam was calculated using the ratio of shear stress to shear rate. The foamed acid was circulated at a constant rate through the coiled loop, where the flow rate was used to calculate the shear rate, and the differential pressure across the coil was used to calculate the shear stress. The viscosity of the foam was modeled using the Power-law model. To generate the required foam quality inside the flow loop, the acid system was displaced by CO₂, while maintaining a back pressure of 1500 psi.

Figure 3 show the CO₂ foam rheological behavior of two Modulated Retarded Acid (MRA) fluid systems was investigated under controlled conditions: a foam quality of approximately 75%, a temperature of 200°F, and a pressure of 1500 psi. Figure 3 (a) chart shows viscosity profile for an acid system without corrosion inhibitor, while the Figure 3 (b) chart shows an acid system with 30 gpt of a corrosion inhibitor. Rheological measurements revealed a significant enhancement in viscosity upon the addition of the corrosion inhibitor. Specifically, the average viscosity of the uninhibited fluid was approximately 80 cP at 100 1/s share rate, whereas the inhibited formulation exhibited a substantially higher viscosity of around 105 cP at 1001/s share rate. Notably, this increase in viscosity was achieved without any observable compromise in foam stability, suggesting that the corrosion inhibitor not only provides corrosion protection but also positively impacts the fluid's rheological performance.

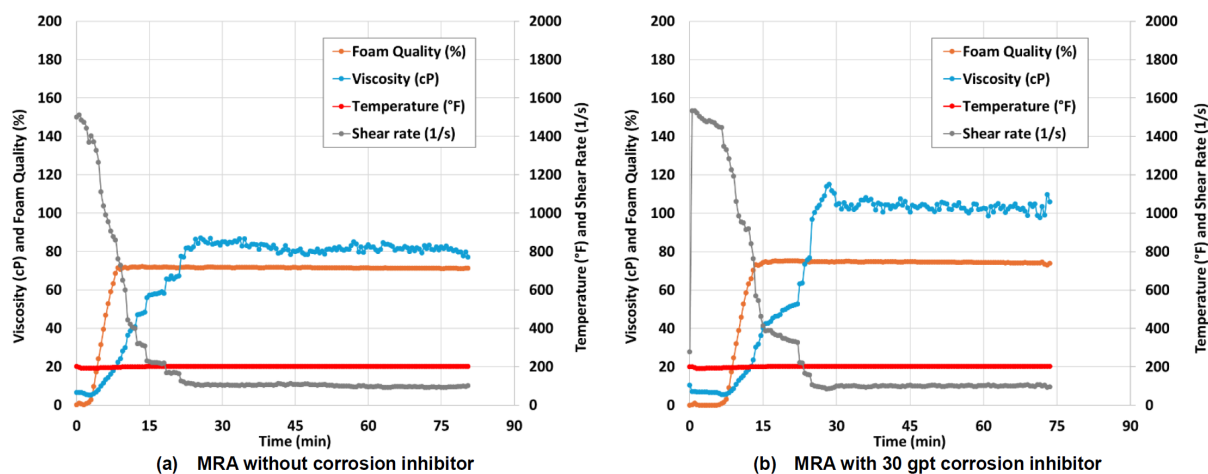


Figure 3—Effect of corrosion inhibitor on CO₂ foam rheology of Modulated Retarded Acid (MRA) at 200F and 1500 psi.

Figure 4 presents a comparison of the CO₂ foamed viscosity of the Modulated Retarded Acid (MRA) fluid system with and without acid retarder. The study was conducted under controlled conditions: a foam quality of 75%, a temperature of 200°F, and a pressure of 1500 psi. The results showed minimal variation, indicating no significant enhancement or degradation in rheological performance. These findings suggest that the retarder has a negligible impact on the viscosity of the MRA-based foam. This outcome is significant for fluid design considerations, as it implies that retarders can be incorporated into the fluid system for their intended chemical function without compromising foam rheology.

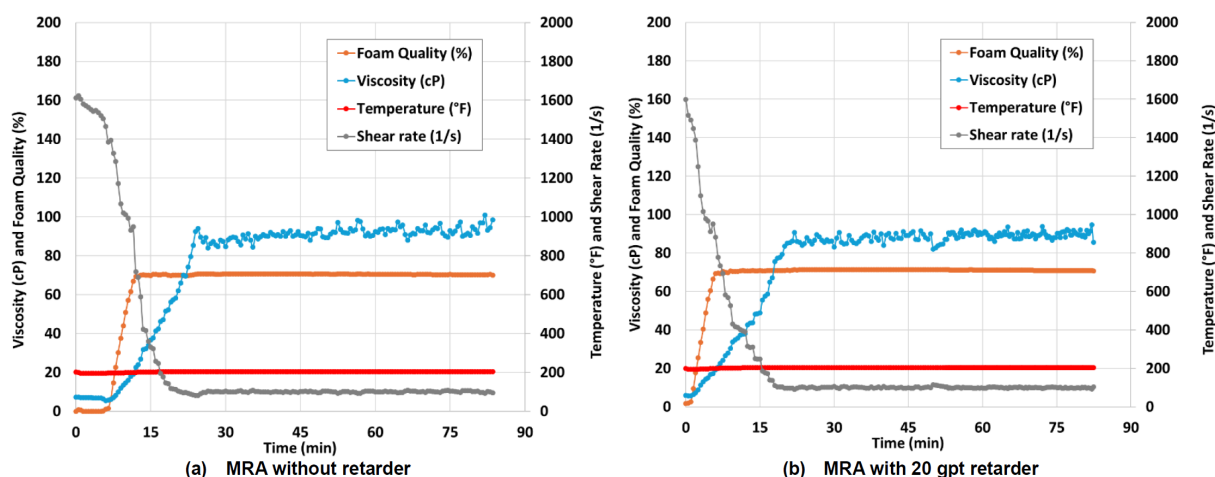


Figure 4—Effect of retarder on CO₂ foam rheology of Modulated Retarded Acid (MRA) at 200°F and 1500 psi.

The Low Viscosity Retarded Acid Systems (LVRA) investigated in this study were prepared with HCl and organic acids for a total acid strength equivalent to 22% HCl which contribute to the retardation of acid reaction with carbonate. Figure 5 presents the CO₂ foam viscosity measured with two different foaming agent concentrations: 10 gpt and 20 gpt. The study was conducted under controlled conditions, including a foam quality of 75%, a temperature of 250°F, and a pressure of 1500 psi. The results show that the higher concentration of foaming agent (20 gpt) yields a viscosity range of 44–45 cP at a shear rate of 100 s⁻¹, similar to that of the 10 gpt foaming agent concentration at 250°F.

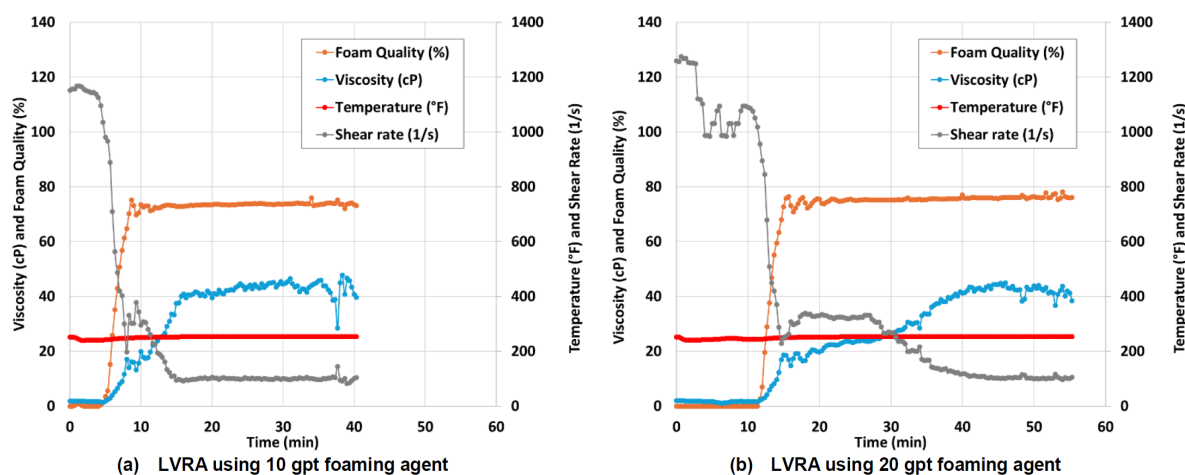


Figure 5—CO₂ foam rheology with increasing foaming agent conc. of Low Viscosity Retarded Acid (LVRA) at 250°F and 1500 psi.

Figure 6 presents a comparison of the CO₂ foamed viscosity of the Low Viscosity Retarded Acid (LVRA) fluid system with and without acid retarder using 10 gpt foaming agent in both tests. The study was conducted under controlled conditions: a foam quality of 75%, a temperature of 250°F, and a pressure of 1500 psi. The CO₂ foamed viscosity was decreased from 44 cP at 100 1/s share rate to 32 cP at 100 1/s shear rate with adding of 2 gpt retarder in LVRA system.

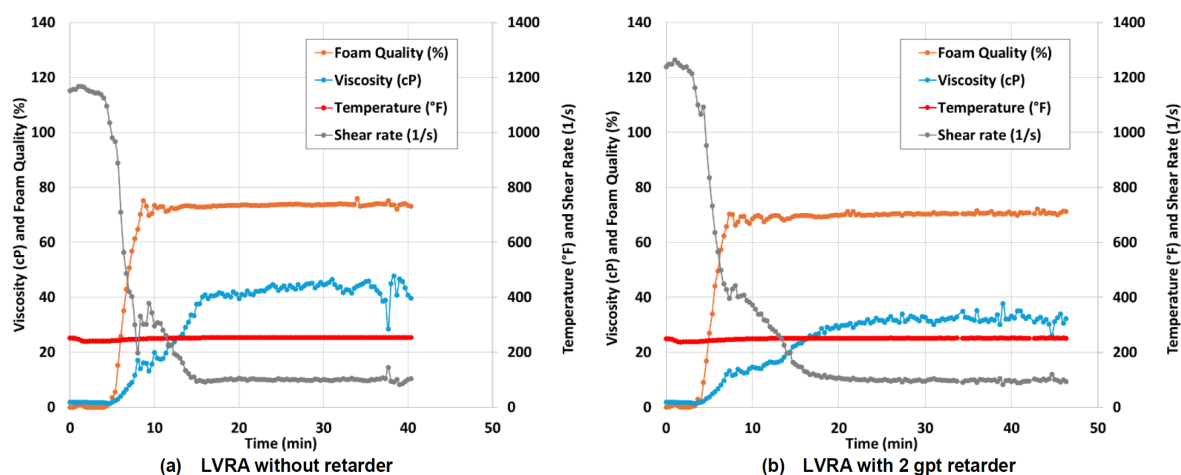


Figure 6—Effect of retarder of CO₂ foam rheology of Rheology of Low Viscosity Retarded Acid (LVRA) at 250°F and 1500 psi.

Conclusions

To enhance flowback recovery, foamed acid fracturing is becoming increasingly important, particularly when the reservoir's energy is insufficient to successfully flow back the well. In addition to its potential properties, such as retardation and deeper conductivity development, foamed acid offers several advantages, including reduced water usage and improved acid diversion. This study develops two novel CO₂ foamed HCl acid systems with retarded acid properties, and based on laboratory investigations, the following conclusions can be drawn:

- The CO₂ foamed rheology of both the MRA and LVRA acid systems exhibited stable viscosity, showing potential for use in acid fracturing applications.
- The development of foam viscosity depends on the viscosity of the external phase fluid. MRA fluid system facilitates the use of any gelling agent and achieves viscosities in the range of 80-105 cP at a shear rate of 100 1/s. In contrast, LVRA, which is designed without a gelling agent, exhibits a viscosity in the range of 44-45 cP at a shear rate of 100 1/s.
- The MRA system exhibited an increase in viscosity with the addition of a corrosion inhibitor, suggesting that the corrosion inhibitor not only provides corrosion protection but also can have a constructive impact on the fluid's rheological performance.
- The CO₂ foamed viscosity of the LVRA system decreased with the addition of a retarder, whereas the MRA system showed no effect from the addition of a retarder.
- The higher concentration of foaming agent (20 gpt) in the LVRA system yields a viscosity range of 44-45 cP at a shear rate of 100 1/s, similar to that obtained with a 10 gpt foaming agent concentration at 250°F.

Both CO₂ foamed acid systems will be further evaluated through high-temperature rheology testing, and the interaction between the foamed acid and carbonate rock will also be investigated using custom-designed acid etching equipment.

Nomenclature

LVRA = low-viscosity retarded acid systems, formulated with different concentrations of hydrochloric acid and organic acids

MRA = modulated retarded acid MRA, formulated with different concentrations of hydrochloric acid and mineral acids

gpt = Gal/1000 Gal

Symbols

V_g = gas volume

V_l = liquid volume

Γ = foam quality

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